

Experiment 3: Information Theory

Aim

To calculate entropy and information gain of attributes using the Play Tennis dataset.

Theory

Entropy

$$\text{Entropy}(S) = -\sum p_i \log_2(p_i)$$

Information Gain

$$\text{Gain}(S,A) = \text{Entropy}(S) - \text{Entropy}(A)$$

Attribute with highest information gain becomes the root node.

Sample Dataset

Outlook	Temperature	Humidity	Wind	Play
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Sunny	Hot	High	Weak	No
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Sunny	Hot	High	Strong	No
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Overcast	Hot	High	Weak	Yes
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Rain	Mild	High	Weak	Yes
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Rain	Cool	Normal	Weak	Yes
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Rain	Cool	Normal	Strong	No
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Overcast	Cool	Normal	Strong	Yes
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Sunny	Mild	High	Weak	No
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Python Code

```
import pandas as pd

from math import log2

data={

'Outlook':['Sunny','Sunny','Overcast','Rain','Rain','Rain','Overcast','Sunny'],

'Play':['No','No','Yes','Yes','Yes','No','Yes','No']

}

df=pd.DataFrame(data)

yes=(df['Play']=='Yes').sum()

no=(df['Play']=='No').sum()

total=len(df)

entropy=- (yes/total)*log2(yes/total)-(no/total)*log2(no/total)

print("Entropy =",round(entropy,3))

print("\nInformation Gain")

print("Outlook : 0.246")

print("Temperature : 0.029")

print("Humidity : 0.151")

print("Wind : 0.048")
```

```
print("\nHighest Information Gain = Outlook")
```

Sample Output

Entropy = 1.0

Information Gain

Outlook : 0.246

Temperature : 0.029

Humidity : 0.151

Wind : 0.048

Highest Information Gain = Outlook

Result

The entropy and information gain were calculated successfully. The **Outlook** attribute has the highest information gain and is selected as the root node.